LECTURE NOTES: SYNTHETIC CATEGORY THEORY

Taught in parallel at Universität Hamburg and Universität Regensburg

LYNE MOSER AND TASHI WALDE

Contents

Introduction 1
What is an ∞-category? 1
Axiomatic approach to ∞-categories 2
Course overview 3
I. Strict category theory 4
1. First definitions 4
1.1. Size issues 4
1.2. Strict categories 5
1.3. Functors between strict categories 6
1.4. Natural transformations between functors 7
2. Yoneda lemma 9
3. Adjunctions and equivalences 11
3.1. Adjunctions 11
3.2. Equivalences of strict categories 12
References 13

Introduction

This course is an introduction to ∞-*category theory*. ∞-Categories are homotopical generalisations of ordinary categories. They provide a convenient language for organising mathematical objects and their relationships, and for formulating and proving theorems. Over the past few decades, they have become an important tool in areas of algebraic topology, algebraic geometry, number theory, homological algebra, and mathematical physics.

What is an ∞-category?

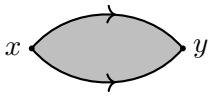
Informally, an ∞-category consists of

- objects $x, y, \ldots$;
- morphisms $f\colon x \to y$ between objects;
- 2-morphisms $x \overset{f}{\underset{g}{\Downarrow \alpha}} y$ between parallel morphisms
- 3-morphisms between parallel 2-morphisms;
- and so on;

such that all k-morphisms for $k \geq 2$ are invertible, in a suitably weak sense. Composition is associative and unital up to higher invertible morphisms, which themselves satisfy compatibility conditions in all higher dimensions.

A motivating example comes from topology. Given a topological space X, consider the ∞-category whose:

- objects are points of X;
- morphisms are paths between points in X; 

- 2-morphisms are homotopies between paths, fixing the endpoints; 

- 3-morphisms are homotopies between homotopies, fixing the boundaries;
- and so on.

Composition is given by concatenating paths and homotopies. Concatenation of paths is associative and unital up to homotopy: different ways of composing the same paths need not give exactly the same parametrised path, but they can be continuously deformed into one another. Moreover, reversing a path gives an inverse up to homotopy. Thus, in this example, even the morphisms are invertible in the appropriate sense. Such a structure is called an ∞-*groupoid*.

Axiomatic approach to ∞-categories

Traditionally, ∞-categories have been studied through concrete *models*, such as quasi-categories [Joy08, Lur09] and complete Segal spaces [Rez01, Ras23]. These models make the informal description above precise using constructions within set-theoretic foundations. Encoding ∞-categories in set-theoretic terms can make them difficult to work with: questions about intrinsic categorical properties may become hard combinatorial problems involving the chosen model.

In this course, we will take an axiomatic approach to ∞-category theory. Our approach differs from the model-based one described above in the following aspects: the foundations, including the structure identity principle, and the synthetic approach.

Homotopy type theory. The first aspect is a change of foundations. In ordinary set-based mathematics, given a set X and elements $x, y \in X$, the expression $x = y$ is a proposition: it is either true or false, and carries no further structure describing an identification between x and y.

In *homotopy type theory*, our basic objects are types. We write $x : X$ to say that x is a term, or element, of the type X. Given terms $x, y : X$, the expression

$$x = y$$

is itself a type, called an *identity type*. Its terms can be thought of as paths identifying x and y. Given two such terms $p, q : x = y$, we can form another identity type

$$p = q,$$

whose terms can be thought of as homotopies between these paths. This process continues indefinitely.

A type therefore already carries higher-dimensional structure: terms, identifications between terms, identifications between identifications, and so on. These identifications behave

like paths and higher homotopies in a space, so types naturally behave as ∞-groupoids. The higher structure is built into our foundations.

The structure identity principle. When studying mathematical structures in classical foundations, one must distinguish between properties that are invariant under isomorphism and those that depend on the chosen presentation. For example, given a group G, the property "G contains an element of order 2" is invariant under group isomorphism, whereas "the identity element of G is the number 0" is not. Indeed, the groups $(\{0,1\}, +\text{mod } 2)$ and $(\{1,-1\}, \cdot)$ are isomorphic, but the latter property holds only for the first. Both properties can be expressed in set-theoretic foundations, but only the first describes the abstract group structure.

An inbuild feature of equality, both in set-theoretic foundations and in homotopy type theory, is the *indiscernibility principle*: if $x = y$, then any property of x also holds for y, and any additional structure on x can be transported to y, and vice-versa.

Homotopy type theory allows for an additional feature that is inconceivable in classical foundations: namely the *structure identity principle*. It connects equality with the mathematical notion of having the same structure. For example, given groups G and H, the terms of the identity type $G = H$ correspond precisely to group isomorphisms from G to H. Thus, giving an identification of two groups amounts to giving an isomorphism between them. The identity type records not just whether the groups can be identified, but also how they are identified.

Combined with indiscernibility, this implies that every property or construction of groups expressible in this framework is invariant under isomorphism. Invariance under isomorphism is therefore built into the foundations.

Synthetic category theory. The second aspect is the synthetic viewpoint itself. Rather than beginning with a concrete construction of an ∞-category, we will specify axioms describing how ∞-categories and the relationships between them behave. Our emphasis will therefore shift from the question "What is an ∞-category?" to "How does the collection of ∞-categories behave?" This requires a sufficiently rich axiomatic setting, but allows us to develop the theory directly from its structural properties.

To do this, we extend homotopy type theory so that we can also talk about non-invertible morphisms. In this framework, ∞-categories are basic types, and we retain the core principles of homotopy type theory, such as the structure identity principle.

Course overview

We will begin with a brief introduction to ordinary category theory, which we will call *strict category theory* throughout these notes. The aim is to become familiar with examples and basic constructions before encountering their ∞-categorical counterparts. We will then survey the foundations of homotopy type theory, following [Rij25, Uni13]. Next, we will pass to an extension of this type theory, which is inspired from [Riv25] and formulate the axioms of synthetic category theory in this extended type theory. This approach to the foundations differs from the one presented in [CCNW21]. We will then develop category theory in this synthetic setting, following [CCNW21].

We plan to cover adjunctions, localisations, left and cocartesian fibrations, the Yoneda lemma, left and right Kan extensions, and limits and colimits. These names need not be familiar at this stage: the course will introduce them progressively, explaining both their meaning and their role.

AI-use disclosure. All mathematical content of these notes is written by the authors. An AI assistant (GPT-6 Astra) was used for proofreading, fixing small errors and typos, and other routine editing tasks.

I. Strict category theory

In this first part, we introduce or recall various definitions and constructions in ordinary category theory that will later be generalized to the ∞-setting. Our aim is to build intuition for the kinds of structures and results we will encounter, so that the abstract setting introduced later feels less intimidating. We will also highlight some of the limitations of strict categories and explain why they can be too rigid to do category theory. For instance, working with *sets* of objects and morphisms encourages us to formulate notions in terms of equality of objects, which can lead to constructions that are not invariant under equivalences of strict categories. Understanding this distinction, and the difficulties it creates, will motivate the synthetic approach to category theory developed later in the course.

1. First definitions

We start by introducing strict categories, functors, and natural transformations, together with some examples. Throughout the course, we refer to ordinary categories as *strict categories*, reserving the term *categories* for their ∞-counterparts, namely synthetic categories, which will play a much more central role in this course.

1.1. Size issues

Russell's paradox tells us that the collection of all sets is not a set. To handle this issue, we will work with three nested universes of sets.

Definition 1.1. A **set-universe** is a set $\mathcal{U}$ whose elements are sets satisfying the following properties:

- (Infinite) the set of natural number $\mathbb{N} \in \mathcal{U}$.
- (Subset closure) if $A \in \mathcal{U}$ and $B \subseteq A$, then $B \in \mathcal{U}$;
- (Power set closure) if $A \in \mathcal{U}$, then its power set $\mathcal{P}(A) \in \mathcal{U}$;
- (Union closure) if $I \in \mathcal{U}$ and $A_i \in \mathcal{U}$, for each $i \in I$, then $\bigcup_{i \in I} A_i \in \mathcal{U}$.

Remark 1.2. A similar proof to Russell's paradox applies here to show that $\mathcal{U} \notin \mathcal{U}$. By contradiction, suppose that $\mathcal{U} \in \mathcal{U}$. Then we can form the subset of $\mathcal{U}$

$$R \coloneqq \{A \in \mathcal{U} \mid A \notin A\} \subseteq \mathcal{U}$$

and, by hypothesis, we have that $R \in \mathcal{U}$. But this gives a contradiction since we have that $R \in R$ if and only if $R \notin R$.

Convention 1.3. We will assume that there exist set-universes

$$\mathcal{U}_{\text{small}} \subseteq \mathcal{U}_{\text{large}} \subseteq \mathcal{U}_{\text{huge}}$$

such that $\mathcal{U}_{\text{small}} \in \mathcal{U}_{\text{large}}$ and $\mathcal{U}_{\text{large}} \in \mathcal{U}_{\text{huge}}$.

Definition 1.4. We introduce the following terminology.

- A **set** or a **small set** is an element of $\mathcal{U}_{\text{small}}$.
- A **large set** is an element of $\mathcal{U}_{\text{large}}$.
- A **huge set** is an element of $\mathcal{U}_{\text{huge}}$.

1.2. Strict categories

Definition 1.5. A **strict category** $\mathcal{C}$ consists of

- a large set $\mathrm{Ob}(\mathcal{C})$ of **objects** $x, y, z, \ldots$;
- for all objects $x, y \in \mathcal{C}$, a large set $\mathcal{C}(x, y)$ of **morphisms** $f\colon x \to y$ from x to y;
- for each object $x \in \mathcal{C}$, an **identity morphism** $\mathrm{id}_x\colon x \to x$;
- for all morphisms $f\colon x \to y$ and $g\colon y \to z$ in $\mathcal{C}$, a **composite morphism** $g \circ f\colon x \to z$; we also write simply gf for the composite;

satisfying the following axioms:

- composition is unital: for each morphism $f\colon x \to y$,
$$f \circ \mathrm{id}_x = f \quad \text{and} \quad \mathrm{id}_y \circ f = f;$$
- composition is associative: for all morphisms $f\colon x \to y$, $g\colon y \to z$, and $h\colon z \to t$ in $\mathcal{C}$,
$$(h \circ g) \circ f = h \circ (g \circ f).$$

Remark 1.6. By convention, a strict category has a large set of objects, which we refer to simply as its **object set**, and large sets of morphisms, which we refer to simply as its **hom sets**.

We will also consider **huge strict categories** with huge sets of objects and of morphisms.

Definition 1.7. A strict category $\mathcal{C}$ is

- **locally small** if, for all objects $x, y \in \mathcal{C}$, the hom set $\mathcal{C}(x, y)$ is small;
- **small** if its object set is small and it is locally small.

Example 1.8. We have the following examples of strict categories.

- The strict category $\mathcal{S}et$ of sets and functions.
- The strict category $\mathcal{S}et_\bullet$ of pointed sets and basepoint-preserving functions.
- For a field k, the strict category $\mathcal{V}ect_k$ of k-vector spaces and k-linear maps.
- The strict category $\mathcal{T}op$ of topological spaces and continuous functions.

All these strict categories are locally small, but not small.

Definition 1.9. Let $\mathcal{C}$ be a strict category. A morphism $f\colon x \to y$ in $\mathcal{C}$ is an **isomorphism** if there is a morphism $g\colon y \to x$ in $\mathcal{C}$ such that $g \circ f = \mathrm{id}_x$ and $f \circ g = \mathrm{id}_y$. We write $f\colon x \xrightarrow{\cong} y$ or $f\colon x \cong y$. We call g an **inverse** of f.

We say that two objects x and y in $\mathcal{C}$ are **isomorphic** if there is an isomorphism $f\colon x \xrightarrow{\cong} y$ in $\mathcal{C}$. We write $x \cong y$.

Example 1.10. The strict categories in Example 1.8 have the following isomorphisms.

- An isomorphism in $\mathcal{S}et$ is a bijection.
- An isomorphism in $\mathcal{S}et_\bullet$ is a basepoint-preserving bijection.
- An isomorphism in $\mathcal{V}ect_k$ is a linear bijection.
- An isomorphism in $\mathcal{T}op$ is a homeomorphism.

Proposition 1.11. *Let $\mathcal{C}$ be a strict category. An isomorphism $f\colon x \to y$ in $\mathcal{C}$ has a unique inverse, which we denote by $f^{-1}\colon y \to x$.*

Proof. Suppose that $g, h\colon y \to x$ are two inverses of f in $\mathcal{C}$. Then
$$g = g \circ \mathrm{id}_y = g \circ (f \circ h) = (g \circ f) \circ h = \mathrm{id}_x \circ h = h. \qquad \square$$

1.3. **Functors between strict categories**

Definition 1.12. Given strict categories $\mathcal{C}$ and $\mathcal{D}$, a **functor** $F\colon \mathcal{C} \to \mathcal{D}$ consists of:

- a function $\mathrm{Ob}(\mathcal{C}) \to \mathrm{Ob}(\mathcal{D})$ sending an object $x \in \mathcal{C}$ to an object $Fx \in \mathcal{D}$;
- for all objects $x, y \in \mathcal{C}$, a function $\mathcal{C}(x,y) \to \mathcal{D}(Fx, Fy)$ sending a morphism $f\colon x \to y$ in $\mathcal{C}$ to a morphism $Ff\colon Fx \to Fy$ in $\mathcal{D}$;

satisfying the following axioms:

- F preserves identities: for each object $x \in \mathcal{C}$,
$$F(\mathrm{id}_x) = \mathrm{id}_{Fx};$$
- F preserves composites: for all morphisms $f\colon x \to y$ and $g\colon y \to z$ in $\mathcal{C}$,
$$F(g \circ f) = Fg \circ Ff.$$

Example 1.13. We have the following examples of functors.

- The forgetful functors $U\colon \mathcal{S}et_\bullet \to \mathcal{S}et$, $U\colon \mathcal{V}ect_k \to \mathcal{S}et$ and $U\colon \mathcal{T}op \to \mathcal{S}et$ sending a pointed set, a k-vector space, or a topological space to its underlying set.
- The functor $(-)_+\colon \mathcal{S}et \to \mathcal{S}et_\bullet$ which freely adjoins a basepoint to a set S, namely $S_+ \coloneqq (S \amalg \{*\}, *)$.
- The functor $\bigoplus_{(-)} k\colon \mathcal{S}et \to \mathcal{V}ect_k$ which sends a set S to the k-vector space with basis S, namely $\bigoplus_S k$.
- The discrete topology functor $D\colon \mathcal{S}et \to \mathcal{T}op$ equipping a set S with the discrete topology, namely $(S, \mathcal{P}(S))$, where $\mathcal{P}(S)$ is the set of subsets of S.
- The indiscrete topology functor $I\colon \mathcal{S}et \to \mathcal{T}op$ equipping a set S with the indiscrete topology, namely $(S, \{\emptyset, S\})$.

Proposition 1.14. *Let $F\colon \mathcal{C} \to \mathcal{D}$ be a functor between strict categories. Then F preserves isomorphisms, i.e., if f is an isomorphism in $\mathcal{C}$, then Ff is an isomorphism in $\mathcal{D}$.*

Proof. Exercise. □

Construction 1.15. There is a strict category $\mathcal{C}at_s$ with

- object set $\mathrm{Ob}(\mathcal{C}at_s)$ consisting of the small strict categories;
- for all objects $\mathcal{C}, \mathcal{D} \in \mathcal{C}at_s$, hom set $\mathcal{C}at_s(\mathcal{C}, \mathcal{D})$ consisting of the functors $\mathcal{C} \to \mathcal{D}$;
- for each object $\mathcal{C} \in \mathcal{C}at_s$, **identity functor** $\mathrm{id}_{\mathcal{C}}\colon \mathcal{C} \to \mathcal{C}$ acting as the identity on objects and on morphisms;
- for all morphisms $F\colon \mathcal{C} \to \mathcal{D}$ and $G\colon \mathcal{D} \to \mathcal{E}$ in $\mathcal{C}at_s$, **composite functor** $GF\colon \mathcal{C} \to \mathcal{E}$ sending
  - an object $x \in \mathcal{C}$ to the object $G(F(x)) \in \mathcal{E}$;
  - a morphism $f\colon x \to y$ in $\mathcal{C}$ to the morphism $G(F(f))\colon G(F(x)) \to G(F(y))$ in $\mathcal{E}$.

This is a well-defined strict category (Exercise).

Similarly, there is a huge strict category $\mathcal{CAT}_s$ of (large) strict categories and functors.

Remark 1.16. The strict category $\mathcal{C}at_s$ of small strict categories and functors defines an object of $\mathcal{CAT}_s$. As for the other examples, the strict category $\mathcal{C}at_s$ is locally small, but not small.

Construction 1.17. Given a strict category $\mathcal{C}$, there is a strict category $\mathcal{C}^{\mathrm{op}}$, called the **opposite strict category** of $\mathcal{C}$, with

- object set $\mathrm{Ob}(\mathcal{C})$;

- for all objects $x, y \in \mathcal{C}$, hom set
$$\mathcal{C}^{\mathrm{op}}(x, y) \coloneqq \mathcal{C}(y, x);$$
given a morphism $f\colon y \to x$ in $\mathcal{C}$, we write $f^{\mathrm{op}}\colon x \to y$ for the corresponding morphism in $\mathcal{C}^{\mathrm{op}}$;
- for each object $x \in \mathcal{C}$, identity morphism $(\mathrm{id}_x)^{\mathrm{op}}$;
- for all morphisms $f^{\mathrm{op}}\colon x \to y$ and $g^{\mathrm{op}}\colon y \to z$ in $\mathcal{C}^{\mathrm{op}}$, composite morphism
$$g^{\mathrm{op}} \circ f^{\mathrm{op}} \coloneqq (f \circ g)^{\mathrm{op}}\colon x \to z.$$

This is a well-defined strict category (Exercise).

Remark 1.18. A **contravariant functor** from $\mathcal{C}$ to $\mathcal{D}$ is a functor $F\colon \mathcal{C}^{\mathrm{op}} \to \mathcal{D}$. It consists of

- a function $\mathrm{Ob}(\mathcal{C}) \to \mathrm{Ob}(\mathcal{D})$ sending an object $x \in \mathcal{C}$ to an object $Fx \in \mathcal{D}$;
- for all objects $x, y \in \mathcal{C}$, a function $\mathcal{C}(x, y) \to \mathcal{D}(Fy, Fx)$ sending a morphism $f\colon x \to y$ in $\mathcal{C}$ to a morphism $Ff\colon Fy \to Fx$ in $\mathcal{D}$;

satisfying the following axioms:

- F preserves identities;
- F preserves compositions: for all morphisms $f\colon x \to y$ and $g\colon y \to z$ in $\mathcal{C}$,
$$F(g \circ f) = Ff \circ Fg.$$

Construction 1.19. Let $\mathcal{C}$ be a locally small strict category and $c \in \mathcal{C}$ be an object. There is a functor
$$\mathcal{C}(-, c)\colon \mathcal{C}^{\mathrm{op}} \to \mathcal{S}et,$$
called the **representable functor** at c, sending:

- an object $x \in \mathcal{C}$ to the (small) set of morphisms $\mathcal{C}(x, c)$;
- a morphism $f\colon x \to y$ in $\mathcal{C}$ to the function
$$f^*\colon \mathcal{C}(y, c) \to \mathcal{C}(x, c), \quad (g\colon y \to c) \mapsto (gf\colon x \to c).$$

This is a well-defined functor (Exercise).

1.4. Natural transformations between functors

Definition 1.20. Given functors $F, G\colon \mathcal{C} \to \mathcal{D}$ between strict categories, a **natural transformation** $\alpha\colon F \Rightarrow G$ consists of:

- for each object $x \in \mathcal{C}$, a morphism $\alpha_x\colon Fx \to Gx$ in $\mathcal{D}$, called the **component** of α at x;

satisfying the following axiom:

- α is natural: for each morphism $f\colon x \to y$ in $\mathcal{C}$, the following square commutes
$$\begin{array}{ccc} Fx & \xrightarrow{\alpha_x} & Gx \\ {\scriptstyle Ff}\downarrow & & \downarrow{\scriptstyle Gf} \\ Fy & \xrightarrow[\alpha_y]{} & Gy \end{array}$$
i.e., $Gf \circ \alpha_x = \alpha_y \circ Ff$.

Construction 1.21. Let $\mathcal{C}$ be a locally small strict category and $f\colon c \to d$ be a morphism in $\mathcal{C}$. There is a natural transformation
$$\mathcal{C}(-, f)\colon \mathcal{C}(-, c) \Rightarrow \mathcal{C}(-, d)$$

whose component at an object $x \in \mathcal{C}$ is the function

$$f_* \colon \mathcal{C}(x,c) \to \mathcal{C}(x,d), \quad (g \colon x \to c) \mapsto (fg \colon x \to d).$$

This is well-defined (Exercise).

Definition 1.22. Given functors $F, G \colon \mathcal{C} \to \mathcal{D}$ between strict categories, a natural transformation $\alpha \colon F \Rightarrow G$ is a **natural isomorphism** if, for each object $x \in \mathcal{C}$, its component $\alpha_x \colon Fx \to Gx$ is an isomorphism in $\mathcal{D}$.

We sometimes say that, for all objects $x \in \mathcal{C}$, there is an isomorphism $Fx \cong Gx$ which is **natural** in x.

Construction 1.23. Given strict categories $\mathcal{C}$ and $\mathcal{D}$, there is a strict category $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$, called the **functor strict category**, with

- object set $\mathrm{Ob}(\mathrm{Fun}(\mathcal{C}, \mathcal{D})) \coloneqq \mathcal{CAT}_s(\mathcal{C}, \mathcal{D})$ consisting of the functors $\mathcal{C} \to \mathcal{D}$;
- for all objects $F, G \in \mathrm{Fun}(\mathcal{C}, \mathcal{D})$, hom set $\mathrm{Fun}(\mathcal{C}, \mathcal{D})(F, G)$ consisting of the natural transformations $F \Rightarrow G$;
- for each object $F \in \mathrm{Fun}(\mathcal{C}, \mathcal{D})$, **identity natural transformation** $\mathrm{id}_F \colon F \Rightarrow F$ having component at an object $c \in \mathcal{C}$ given by
$$(\mathrm{id}_F)_c \coloneqq \mathrm{id}_{Fc} \colon Fc \to Fc;$$
- for all morphisms $\alpha \colon F \Rightarrow G$ and $\beta \colon G \Rightarrow H$ in $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$, **composite natural transformation** $\beta \circ \alpha \colon F \Rightarrow H$ having component at an object $c \in \mathcal{C}$ given by the composite
$$(\beta \circ \alpha)_c \coloneqq \beta_c \circ \alpha_c \colon Fc \to Hc.$$

This is a well-defined strict category (Exercise).

Proposition 1.24. *Given strict categories $\mathcal{C}$ and $\mathcal{D}$, a natural transformation is an isomorphism in $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$ if and only if it is a natural isomorphism.*

Proof. Exercise. □

Construction 1.25. The **terminal strict category** is the strict category $[0]$ with a single object $*$ and a single morphism id_*.

Proposition 1.26. *Given a strict category $\mathcal{C}$, there is an isomorphism of strict categories*

$$\mathrm{Fun}([0], \mathcal{C}) \cong \mathcal{C}$$

which is natural in $\mathcal{C}$.

Proof. Exercise. □

Remark 1.27. As a consequence of this result, we have that an object $c \in \mathcal{C}$ corresponds to a unique functor $c \colon [0] \to \mathcal{C}$, which we also call c.

Construction 1.28. Given a functor $F \colon \mathcal{C} \to \mathcal{D}$ between strict categories and a strict category $\mathcal{E}$, there is a functor

$$F^* \colon \mathrm{Fun}(\mathcal{D}, \mathcal{E}) \to \mathrm{Fun}(\mathcal{C}, \mathcal{E}),$$

called the **precomposition functor** with F, sending

- a functor $G \colon \mathcal{D} \to \mathcal{E}$ to the composite $GF \colon \mathcal{C} \to \mathcal{E}$;
- a natural transformation $\alpha \colon G \Rightarrow H$ between functors $G, H \colon \mathcal{D} \to \mathcal{E}$ to the natural transformation $\alpha_F \colon GF \Rightarrow HF$ whose component at an object $x \in \mathcal{C}$ is the morphism in $\mathcal{E}$
$$\alpha_{Fx} \colon G(Fx) \to H(Fx).$$

There is also a functor

$$F_*\colon \mathrm{Fun}(\mathcal{E},\mathcal{C}) \to \mathrm{Fun}(\mathcal{E},\mathcal{D}),$$

called the **postcomposition functor** with F, sending

- a functor $G\colon \mathcal{E} \to \mathcal{C}$ to the composite $FG\colon \mathcal{E} \to \mathcal{D}$;
- a natural transformation $\alpha\colon G \Rightarrow H$ between functors $G, H\colon \mathcal{E} \to \mathcal{C}$ to the natural transformation $F\alpha\colon FG \Rightarrow FH$ whose component at an object $x \in \mathcal{E}$ is the morphism in $\mathcal{D}$

$$F(\alpha_x)\colon F(Gx) \to F(Hx).$$

These are well-defined functors (Exercise).

2. Yoneda lemma

We now turn to one of the fundamental results of category theory, the *Yoneda lemma*. An important consequence of this result is that an object in a strict category is determined, up to isomorphism, by its representable functor. Hence we can understand an object (up to isomorphism) through its relationships with other objects. It also highlights the central role of the strict category $\mathcal{S}et$ in strict category theory. For instance, we can detect isomorphisms in a strict category by proving that certain functions of sets are bijections.

Theorem 2.1 (Yoneda lemma)**.** *Let $\mathcal{C}$ be a locally small strict category. For all objects $c \in \mathcal{C}$ and functors $F\colon \mathcal{C}^{\mathrm{op}} \to \mathcal{S}et$, there is a bijection of sets*

$$\mathrm{Fun}(\mathcal{C}^{\mathrm{op}},\mathcal{S}et)(\mathcal{C}(-,c),F) \cong Fc$$

which is natural in both c and F.

Proof. Evaluation at the identity morphism defines a function

$$\mathrm{Fun}(\mathcal{C}^{\mathrm{op}},\mathcal{S}et)(\mathcal{C}(-,c),F) \to Fc, \quad (\alpha\colon \mathcal{C}(-,c) \Rightarrow F) \mapsto \alpha_c(\mathrm{id}_c),$$

where $\alpha_c\colon \mathcal{C}(c,c) \to Fc$ is the function given by the component of α at c.

Next, consider the function

$$Fc \to \mathrm{Fun}(\mathcal{C}^{\mathrm{op}},\mathcal{S}et)(\mathcal{C}(-,c),F), \quad u \mapsto \alpha_u$$

sending an element $u \in Fc$ to $\alpha_u\colon \mathcal{C}(-,c) \Rightarrow F$ with component at an object $x \in \mathcal{C}$ given by the function

$$(\alpha_u)_x\colon \mathcal{C}(x,c) \to Fx, \quad (f\colon x \to c) \mapsto Ff(u),$$

where $Ff\colon Fc \to Fx$ is the function given by the image of f under F. This is a well-defined natural transformation (Exercise).

These functions are mutually inverse and natural in c and F (Exercise). □

Construction 2.2. Let $\mathcal{C}$ be a locally small strict category. There is a functor

$$\mathbf{y}_{\mathcal{C}}\colon \mathcal{C} \to \mathrm{Fun}(\mathcal{C}^{\mathrm{op}},\mathcal{S}et)$$

called the **Yoneda functor**, sending

- an object $c \in \mathcal{C}$ to the representable functor $\mathcal{C}(-,c)\colon \mathcal{C}^{\mathrm{op}} \to \mathcal{S}et$;
- a morphism $f\colon c \to d$ in $\mathcal{C}$ to the natural transformation $\mathcal{C}(-,f)\colon \mathcal{C}(-,c) \Rightarrow \mathcal{C}(-,d)$.

This is well-defined (Exercise).

Definition 2.3. A functor $F\colon \mathcal{C} \to \mathcal{D}$ between strict categories is **fully faithful** if, for all objects $x, y \in \mathcal{C}$, the induced function

$$\mathcal{C}(x,y) \to \mathcal{D}(Fx,Fy)$$

is a bijection of sets.

Corollary 2.4. *Let $\mathcal{C}$ be a locally small strict category. The Yoneda functor*

$$\mathbf{y}_{\mathcal{C}} \colon \mathcal{C} \to \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S}et)$$

is fully faithful.

Proof. Exercise. □

Proposition 2.5. *Let $F \colon \mathcal{C} \to \mathcal{D}$ be a fully faithful functor. Then F reflects isomorphisms, i.e., if f is a morphism in $\mathcal{C}$ such that Ff is an isomorphism in $\mathcal{D}$, then f is an isomorphism in $\mathcal{C}$.*

Moreover, two objects $x, y \in \mathcal{C}$ are isomorphic if and only if their images $Fx, Fy \in \mathcal{D}$ are isomorphic.

Proof. Exercise. □

Corollary 2.6. *Let $\mathcal{C}$ be a locally small strict category. The following are equivalent for a morphism $f \colon c \to d$ in $\mathcal{C}$.*

(i) The morphism f is an isomorphism in $\mathcal{C}$.
(ii) For all objects $x \in \mathcal{C}$, the induced function

$$f_* \colon \mathcal{C}(x, c) \to \mathcal{C}(x, d)$$

is a bijection of sets.
(iii) For all objects $x \in \mathcal{C}$, the induced function

$$f^* \colon \mathcal{C}(d, x) \to \mathcal{C}(c, x)$$

is a bijection of sets.

Proof. Since the Yoneda functor $\mathbf{y}_{\mathcal{C}} \colon \mathcal{C} \to \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S}et)$ is fully faithful, it preserves and reflects isomorphisms. Hence the morphism $f \colon c \to d$ is an isomorphism in $\mathcal{C}$ if and only if the natural transformation $\mathcal{C}(-, f) \colon \mathcal{C}(-, c) \Rightarrow \mathcal{C}(-, d)$ is an isomorphism in $\mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S}et)$. By Proposition 1.24, this holds if and only if it is a natural isomorphism, i.e., for every object $x \in \mathcal{C}$, the induced morphism

$$f_* \colon \mathcal{C}(x, c) \to \mathcal{C}(x, d)$$

is an isomorphism in $\mathcal{S}et$. This proves that (i) and (ii) are equivalent.

Next, observe that f is an isomorphism in $\mathcal{C}$ if and only if $f^{\mathrm{op}} \colon d \to c$ is an isomorphism in $\mathcal{C}^{\mathrm{op}}$ (Exercise). Then the equivalence between (i) and (ii) for $\mathcal{C}^{\mathrm{op}}$ implies that f^{op} is an isomorphism in $\mathcal{C}^{\mathrm{op}}$ if and only if, for every object $x \in \mathcal{C}$, the induced morphism

$$f^{\mathrm{op}}_* = f^* \colon \mathcal{C}^{\mathrm{op}}(x, d) = \mathcal{C}(d, x) \to \mathcal{C}(c, x) = \mathcal{C}^{\mathrm{op}}(x, c)$$

is an isomorphism in $\mathcal{S}et$. This proves that (i) and (iii) are equivalent. □

Corollary 2.7. *Let $\mathcal{C}$ be a locally small strict category. Two objects $x, y \in \mathcal{C}$ are isomorphic if and only if their representable functors $\mathcal{C}(-, x)$ and $\mathcal{C}(-, y)$ are naturally isomorphic.*

Proof. This follows from the preceding proposition applied to the fully faithful Yoneda functor, together with Proposition 1.24. □

3. Adjunctions and equivalences

We now turn to two important concepts in strict category theory, namely *adjunctions* and *equivalences*. Adjunctions provide a way to describe universal constructions and organize them into functors. Equivalences express when two strict categories should be thought of as being the "same".

From a categorical point of view, two isomorphic objects should be interchangeable, even if they are not equal. However, an isomorphism of strict categories requires a bijection between the sets of objects, making it too restrictive for this purpose. Equivalences provide a weakening of the notion of isomorphism between strict categories and capture the intended notion of sameness: they detect objects only up to isomorphism, rather than up to equality.

A fundamental difficulty with the set-theoretic description of strict categories is that equality and isomorphism of objects remain distinct notions. Consequently, properties and constructions can depend on literal equalities between objects and fail to be well-behaved with respect to isomorphisms. We will illustrate this with examples of properties and constructions that are not invariant under replacing objects by isomorphic ones or strict categories by equivalent ones. Further examples will appear in the following sections.

The synthetic approach will address this foundational issue by not allowing a distinction between equality and isomorphism of objects, so that invariance under isomorphism is built into the theory. Similarly, the appropriate notion of isomorphism between synthetic categories will generalize that of an equivalence between strict categories.

3.1. **Adjunctions**

Definition 3.1. Given strict categories $\mathcal{C}$ and $\mathcal{D}$, an **adjunction** between $\mathcal{C}$ and $\mathcal{D}$ consists of:

- a functor $L\colon \mathcal{C} \to \mathcal{D}$, called the **left adjoint**;
- a functor $R\colon \mathcal{D} \to \mathcal{C}$, called the **right adjoint**;
- a natural transformation $\eta\colon \mathrm{id}_{\mathcal{C}} \Rightarrow RL$, called the **unit**;
- a natural transformation $\varepsilon\colon LR \Rightarrow \mathrm{id}_{\mathcal{D}}$, called the **counit**;

such that, for all objects $x \in \mathcal{C}$ and $y \in \mathcal{D}$, the following triangles commute in $\mathcal{D}$ and $\mathcal{C}$, respectively:



We write $L\colon \mathcal{C} \rightleftarrows \mathcal{D} : R$ or $L \dashv R$ for such an adjunction.

Example 3.2. We have the following examples of adjunctions between strict categories.

- The free-forgetful adjunctions

$$(-)_+\colon \mathcal{S}et \rightleftarrows \mathcal{S}et_\bullet : U \quad \text{and} \quad \bigoplus_{(-)} k\colon \mathcal{S}et \rightleftarrows \mathcal{V}ect_k : U.$$

- The discrete-forgetful and forgetful-indiscrete adjunctions

$$D\colon \mathcal{S}et \rightleftarrows \mathcal{T}op : U \quad \text{and} \quad U\colon \mathcal{T}op \rightleftarrows \mathcal{S}et : I.$$

See Example 1.13 for the definition of the functors involved.

Theorem 3.3 (Adjunctions representably)**.** *Given functors $L\colon \mathcal{C} \to \mathcal{D}$ and $R\colon \mathcal{D} \to \mathcal{C}$ between strict categories, there is a unit and a counit making the pair (L, R) into an adjunction $L \dashv R$ if and only if, for all objects $x \in \mathcal{C}$ and $y \in \mathcal{D}$, there is a bijection*

$$\mathcal{D}(Lx, y) \cong \mathcal{C}(x, Ry)$$

which is natural in x and y.

Proof. Given an adjunction $L \dashv R$, the unit $\eta\colon \mathrm{id}_{\mathcal{C}} \Rightarrow RL$ induces a function

$$\mathcal{D}(Lx, y) \xrightarrow{R} \mathcal{C}(RLx, Ry) \xrightarrow{(\eta_x)^*} \mathcal{C}(x, Ry), \quad (g\colon Lx \to y) \mapsto R(g) \circ \eta_x,$$

and the counit $\varepsilon\colon LR \Rightarrow \mathrm{id}_{\mathcal{D}}$ induces a function

$$\mathcal{C}(x, Ry) \xrightarrow{L} \mathcal{D}(Lx, LRy) \xrightarrow{(\varepsilon_y)_*} \mathcal{D}(Lx, y), \quad (f\colon x \to Ry) \mapsto \varepsilon_y \circ L(f).$$

These are mutually inverse and natural in x and y (Exercise).

Conversely, suppose that, for all objects $x \in \mathcal{C}$ and $y \in \mathcal{D}$, we are given natural bijections

$$\varphi_{x,y}\colon \mathcal{D}(Lx, y) \cong \mathcal{C}(x, Ry).$$

Define $\eta\colon \mathrm{id}_{\mathcal{C}} \Rightarrow RL$ whose component at an object $x \in \mathcal{C}$ is given by

$$\eta_x \coloneqq \varphi_{x,Lx}(\mathrm{id}_{Lx})\colon x \to RLx.$$

Define $\varepsilon\colon LR \Rightarrow \mathrm{id}_{\mathcal{D}}$ whose component at an object $y \in \mathcal{D}$ is given by the morphism

$$\varepsilon_y \coloneqq \varphi^{-1}_{Ry,y}(\mathrm{id}_{Ry})\colon LRy \to y.$$

These are well-defined natural transformations satisfying the triangle identities (Exercise). This proves that $L \dashv R$ is an adjunction. □

3.2. **Equivalences of strict categories**

Recall that a functor $F\colon \mathcal{C} \to \mathcal{D}$ between strict categories is an isomorphism in $\mathcal{CAT}_s$ if there is a functor $G\colon \mathcal{D} \to \mathcal{C}$ such that $GF = \mathrm{id}_{\mathcal{C}}$ and $FG = \mathrm{id}_{\mathcal{D}}$.

Since we have a notion of morphisms between functors, namely the natural transformations, we can weaken the notion of sameness between strict categories as follows.

Definition 3.4. A functor $F\colon \mathcal{C} \to \mathcal{D}$ between strict categories is an **equivalence** if there is a functor $G\colon \mathcal{D} \to \mathcal{C}$ and natural isomorphisms $\eta\colon \mathrm{id}_{\mathcal{C}} \overset{\cong}{\Longrightarrow} GF$ and $\varepsilon\colon FG \overset{\cong}{\Longrightarrow} \mathrm{id}_{\mathcal{D}}$.

Example 3.5. The following classes of functors give special cases of equivalences.

- Any isomorphism in $\mathcal{CAT}_s$ is an equivalence of strict categories.
- Given an adjunction $L \dashv R$ such that the unit and the counit are natural isomorphisms, then both L and R are equivalences of strict categories; we refer to this as an **adjoint equivalence**.

Fact: every equivalence of strict categories can be promoted to an adjoint equivalence; see [Rie16, Corollary 4.3.6]. We won't use this fact in this course.

Definition 3.6. A functor $F\colon \mathcal{C} \to \mathcal{D}$ between strict categories is **essentially surjective** if, for all objects $d \in \mathcal{D}$, there is an object $c \in \mathcal{C}$ and an isomorphism $\varphi\colon Fc \xrightarrow{\cong} d$ in $\mathcal{D}$.

Theorem 3.7 (Fundamental Theorem)**.** *A functor $F\colon \mathcal{C} \to \mathcal{D}$ between strict categories is an equivalence if and only if it is essentially surjective on objects and fully faithful.*

Proof. Next time. □

Construction 3.8. The **walking isomorphism** is the strict category $\mathcal{I} \coloneqq \{0 \cong 1\}$ with two objects 0 and 1, whose only morphisms are the identities and mutually inverse isomorphisms $f\colon 0 \to 1$ and $f^{-1}\colon 1 \to 0$. Observe that the hom sets of $\mathcal{I}$ are all singletons.

Proposition 3.9. *The functor* $0\colon [0] \to \mathcal{I}$ *is an equivalence of strict categories.*

Proof. It is surjective on objects up to isomorphisms, since $0 \cong 0$ in $\mathcal{I}$, and it is fully faithful since, for the unique object $* \in [0]$, we have a bijection

$$[0](*,*) = \{\mathrm{id}_*\} \xrightarrow{\cong} \{\mathrm{id}_0\} = \mathcal{I}(0,0). \qquad \square$$

References

[CCNW21] Bastiaan Cnossen, Denis-Charles Cisinski, Kim Nguyen, and Tashi Walde, *Synthetic category theory*, available here, 2021.

[Joy08] André Joyal, *The theory of quasi-categories and its applications*, available here, 2008.

[Lur09] Jacob Lurie, *Higher topos theory*, Annals of Mathematics Studies, vol. 170, Princeton University Press, Princeton, NJ, 2009. MR 2522659

[Ras23] Nima Rasekh, *Yoneda Lemma for Simplicial Spaces*, Appl. Categ. Structures **31** (2023), no. 4, 27. MR 4616631

[Rez01] Charles Rezk, *A model for the homotopy theory of homotopy theory*, Trans. Amer. Math. Soc. **353** (2001), no. 3, 973–1007. MR 1804411

[Rie16] Emily Riehl, *Category theory in context*, Aurora Dover Modern Math Originals, Dover Publications, Inc., Mineola, NY, 2016. MR 4727501

[Rij25] Egbert Rijke, *Introduction to homotopy type theory*, Cambridge Studies in Advanced Mathematics, vol. 219, Cambridge University Press, 2025.

[Riv25] Michele Riva, *A type theory for synthetic categories*, available here, 2025.

[Uni13] The Univalent Foundations Program, *Homotopy type theory: Univalent foundations of mathematics*, available here, 2013.

Fachbereich Mathematik, Universität Hamburg, Hamburg, Germany
Email address: lyne.moser@uni-hamburg.de

Fakultät für Mathematik, Universität Regensburg, Regensburg, Germany
Email address: tashi.walde@ur.de